

FULL STACK DEVELOPMENT - I (FSD-1)

Unit 2: HTML & Web Essentials (Sr. No. 88 to 107)

FOR REFERENCE ONLY

Question 88: Write HTML tags to generate the given output.

Key Concept & Explanation: Subscripts are rendered using <sub> and superscripts using <sup>. Pure HTML uses native formatting tags (, <sup>, <sub>) and table attributes (border, cellpadding, cellspacing, align, rowspan, colspan) without any CSS.

Pure HTML Code for Expressions:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mathematical and Chemical Expressions</title>
</head>
<body>
  <h3>Mathematical Expression:</h3>
  <p><b>(A + B)<sup>2</sup></b> = A<sup>2</sup> + 2AB + B<sup>2</sup></p>

  <h3>Chemical Expression:</h3>
  <p><b>(C<sub>2</sub>H<sub>3</sub>O<sub>2</sub>)<sup>2</sup></b></p>
</body>
</html>
```

Pure HTML Code for Irregular Table:

```
<!DOCTYPE html>
<html>
<head>
  <title>Irregular Table</title>
</head>
<body>
  <table border="1" cellpadding="10" cellspacing="0" align="center">
    <tr align="center">
      <td rowspan="3">A</td>
      <td>B</td>
      <td colspan="2">C</td>
      <td>D</td>
      <td rowspan="2" colspan="2">E</td>
    </tr>
    <tr align="center">
      <td>F</td>
      <td>G</td>
      <td rowspan="3">H</td>
    </tr>
  </table>
```

```

        <td>I</td>
    </tr>
    <tr align="center">
        <td rowspan="2">J</td>
        <td>K</td>

        <td>L</td>
        <td>M</td>
        <td>N</td>
    </tr>
    <tr align="center">
        <td>O</td>
        <td>P</td>
        <td colspan="3">Q</td>
    </tr>
</table>
</body>
</html>

```

Question 89: Give the code to print given table.

Key Concept & Explanation: Pure HTML table using native table attributes: border, cellpadding, cellspacing, rowspan, and colspan without any CSS.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>Average Table</title>
</head>
<body>
    <table border="1" cellpadding="8" cellspacing="0" align="center">
        <tr align="center">
            <th rowspan="2"></th>
            <th colspan="2">average</th>
            <th rowspan="2">other<br>category</th>
        </tr>
        <tr align="center">
            <th>height</th>
            <th>weight</th>
        </tr>
        <tr align="center">
            <th>males</th>
            <td>1.9</td>
            <td>0.003</td>
            <td>yyy</td>
        </tr>
        <tr align="center">
            <th>females</th>
            <td>1.7</td>

```

```

        <td>0.002</td>
        <td>xxx</td>
    </tr>
</table>
</body>
</html>

```

Question 90: Write an HTML script to code given table:

Key Concept & Explanation: Pure HTML supports the native 'bgcolor' attribute directly on <td> and <tr> elements, allowing colors without any external or internal CSS.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>Table with Cell Spanning and BGColor</title>
</head>
<body>
    <table border="1" cellpadding="12" cellspacing="0" width="100%" align="center">
        <caption><b><font size="5">Table</font></b></caption>
        <!-- Row 1 -->
        <tr align="center">
            <td rowspan="2" bgcolor="yellow"><b>A</b></td>
            <td bgcolor="lightgreen"><b>B</b></td>
            <td rowspan="3" bgcolor="yellow"><b>D</b></td>
            <td colspan="2" bgcolor="pink"><b>E</b></td>
            <td bgcolor="lightblue"><b>F</b></td>
        </tr>
        <!-- Row 2 -->
        <tr align="center">
            <td bgcolor="lightblue"><b>C</b></td>
            <td rowspan="2" bgcolor="lightblue"><b>G</b></td>
            <td rowspan="2" bgcolor="lightgreen"><b>H</b></td>
            <td rowspan="2" bgcolor="yellow"><b>I</b></td>
        </tr>
        <!-- Row 3 -->
        <tr align="center">
            <td colspan="2" bgcolor="pink"><b>J</b></td>
        </tr>
        <!-- Row 4 -->
        <tr align="center">
            <td colspan="3" bgcolor="lightgreen"><b>K</b></td>
            <td bgcolor="yellow"><b>L</b></td>
            <td colspan="2" bgcolor="pink"><b>M</b></td>
        </tr>
    </table>
</body>
</html>

```

Question 91: Draw table as asked.

Key Concept & Explanation: Constructed using pure HTML attributes: 'border', 'cellpadding', 'cellspacing', 'align', 'width', 'rowspan', and 'colspan'.

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Purchased Equipments</title>
</head>
<body>
  <table border="1" cellpadding="6" cellspacing="5" align="center">
    <!-- Top Title Row -->
    <tr align="center">
      <th colspan="4">Purchased Equipments (June, 2006)</th>
    </tr>
    <!-- Headers -->
    <tr align="center">
      <th rowspan="2">Item Num#</th>
      <th rowspan="2">Item Picture</th>
      <th>Item Description</th>
      <th>Price</th>
    </tr>
    <tr align="center">
      <th>Shipping Handling, Installation, etc</th>
      <th>Expense</th>
    </tr>
    <!-- Item 1 -->
    <tr>
      <td rowspan="2" align="center">1.</td>
      <td rowspan="2" align="center">
        
      </td>
      <td>IBM Clone Computer.</td>
      <td align="center">$ 400.00</td>
    </tr>
    <tr>
      <td>Shipping Handling, Installation, etc</td>
      <td align="center">$ 20.00</td>
    </tr>
    <!-- Item 2 -->
    <tr>
      <td rowspan="2" align="center">2.</td>
      <td rowspan="2" align="center">
        
      </td>
      <td>1GB RAM Module for Computer.</td>
      <td align="center">$ 50.00</td>
    </tr>
    <tr>
      <td>Shipping Handling, Installation, etc</td>
```

```

        <td align="center">$ 14.00</td>
    </tr>
    <!-- Bottom Summary Row -->
    <tr align="center">
        <th colspan="4">Purchased Equipments (June, 2006)</th>
    </tr>
</table>
</body>
</html>

```

Question 92: Write an HTML script to play a YouTube video looping forever & auto-playing while page load.

Key Concept & Explanation: Autoplay and looping are controlled strictly via YouTube iframe URL parameters: 'autoplay=1', 'mute=1' (required by browser autoplay policy), 'loop=1', and 'playlist=VIDEO_ID'. Standard HTML iframe attributes 'width', 'height', 'src', 'frameborder', and 'allowfullscreen' are used.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>YouTube Video Autoplay and Loop</title>
</head>
<body>
    <h2>YouTube Video - Autoplay & Loop Forever</h2>
    <iframe
        width="560"
        height="315"

src="https://www.youtube.com/embed/dQw4w9WgXcQ?autoplay=1&mute=1&loop=1&playlist=dQw4w9WgXcQ"
        allowfullscreen>
    </iframe>
</body>
</html>

```

Question 93: Write HTML code to embed a audio in a web document.

Key Concept & Explanation: Pure HTML5 provides the native <audio> tag with attributes: 'controls' (displays native audio controls), 'autoplay', 'muted', and 'loop', alongside nested <source> tags.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>Audio Embedding</title>
</head>
<body>
    <h2>Pure HTML5 Audio Player</h2>

```

```

<audio controls autoplay muted loop>
  <source src="song.mp3" type="audio/mpeg">
  <source src="song.ogg" type="audio/ogg">
  Your browser does not support the audio element.
</audio>
</body>
</html>

```

Question 94: Write HTML code to embed a video in a web document.

Key Concept & Explanation: Native HTML5 <video> tag with attributes: 'width', 'height', 'controls', 'poster', and 'preload', with nested <source> elements.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Video Embedding</title>
</head>
<body>
  <h2>Pure HTML5 Video Player</h2>
  <video width="640" height="360" controls poster="thumbnail.jpg">
    <source src="movie.mp4" type="video/mp4">
    <source src="movie.webm" type="video/webm">
    Your browser does not support the video tag.
  </video>
</body>
</html>

```

Question 95: Write HTML code to play your YouTube video loop forever.

Key Concept & Explanation: Uses standard HTML <iframe> with the video ID repeated in the 'playlist' URL parameter to achieve infinite looping.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Looping YouTube Video</title>
</head>
<body>
  <h2>YouTube Infinite Loop Player</h2>
  <iframe
    width="640"
    height="360"
    src="https://www.youtube.com/embed/tgbNymZ7vqY?loop=1&playlist=tgbNymZ7vqY"
    frameborder="0"

```

```
        allowfullscreen>
    </iframe>
</body>
</html>
```

Question 96: Write HTML code that includes aside & article tags.

Key Concept & Explanation: HTML5 semantic elements structure document layout natively without requiring any CSS classes. <article> represents self-contained content, and <aside> represents tangential sidebar content.

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Semantic Tags: Article and Aside</title>
</head>
<body>
    <article>
        <h1>Understanding HTML5 Semantics</h1>
        <p>HTML5 introduces elements that clearly describe their meaning to both the browser and
the developer.</p>
        <p>Examples of semantic elements include header, footer, section, article, and aside.</p>
    </article>

    <aside>
        <h3>Related Information</h3>
        <p>Semantic tags improve web accessibility and search engine ranking.</p>
        <ul>
            <li><a href="#html5">HTML5 Guide</a></li>
            <li><a href="#tags">List of Tags</a></li>
        </ul>
    </aside>
</body>
</html>
```

Question 97: Write HTML code having article title and two sections in article by using <article> and <aside> tag.

Key Concept & Explanation: A single <article> enclosing an <h1> heading, two distinct <section> elements, and a complementary <aside> block, using pure HTML structure.

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Article with Sections and Aside</title>
</head>
```

```

<body>
  <article>
    <!-- Article Title -->
    <h1>Full-Stack Web Development Architecture</h1>

    <!-- Section 1 -->
    <section>
      <h2>Section 1: Frontend Technologies</h2>
      <p>The client side handles presentation and user interaction using HTML, forms,
tables, and media elements.</p>
    </section>

    <!-- Section 2 -->
    <section>
      <h2>Section 2: Backend Technologies</h2>
      <p>The server side handles application logic, user authentication, and communication
with persistent databases.</p>
    </section>

    <!-- Aside inside/alongside the article -->
    <aside>
      <h3>Quick Note</h3>
      <p>HTTP and RESTful APIs connect the client and server together.</p>
    </aside>
  </article>
</body>
</html>

```

Question 98: Differentiate between <div> and tags.

Key Concept & Explanation: Both are generic container elements in pure HTML, distinguished by their default display behavior:

Comparison Point	<div> Tag	 Tag
Element Category	Block-level element	Inline element
Line Breaking	Always begins on a new line and spans the full available width.	Does not start on a new line; occupies only the space required by its content.
Permitted Content	Can contain other block-level elements, paragraphs, tables, lists, and inline tags.	Can only contain text and other inline tags; cannot nest block-level elements.
Primary Purpose	Grouping larger sections, structural blocks, or layouts.	Grouping words, phrases, or inline text characters for formatting.
HTML Syntax Example	Content	<p>This is a highlighted

Question 99: Create a simple HTML webpage that demonstrates the use of the <iframe> tag to embed different types of resources. The page should include:

- 1) A PDF document embedded using an iframe.
- 2) A Google Map embedded using an iframe.
- 3) A local HTML file embedded within the main page using an iframe.

Key Concept & Explanation: Demonstrates embedding three distinct resource types using purely native <iframe> HTML attributes: 'src', 'width', 'height', 'frameborder', and fallback anchor tags.

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Iframe Resource Demonstrations</title>
</head>
<body>
  <h1>Demonstrating Multiple Resources Using Iframe</h1>

  <!-- 1) Embedded PDF Document -->
  <h3>1. Embedded PDF Document:</h3>
  <iframe src="sample.pdf" width="100%" height="300" frameborder="1">
    <p>Your browser does not support iframes. <a href="sample.pdf">Download PDF</a></p>
  </iframe>

  <!-- 2) Embedded Google Map -->
  <h3>2. Embedded Google Map:</h3>
  <iframe
src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d3671.942918840292!2d72.571362!3d23.0225
05!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1!3m3!1m2!1s0x0%3A0x0!2zMjPCsDAXJzIxLjAiTiA3MsKwMzQnMTYuO
SJF!5e0!3m2!1sen!2sin!4v1600000000000!5m2!1sen!2sin"
width="600"
height="300"
frameborder="1"
allowfullscreen>
  </iframe>

  <!-- 3) Embedded Local HTML Page -->
  <h3>3. Embedded Local HTML File:</h3>
  <iframe src="inner_page.html" width="100%" height="200" frameborder="1">
    <p>Your browser does not support iframes.</p>
  </iframe>
</body>
</html>
```

Question 100: How <th>, <thead> and <caption> tags differ?

Key Concept & Explanation: These three elements define different structural tiers of an HTML <table>:

Tag	Role / Meaning	Default Browser Appearance	Placement Rule
<caption>	Provides the title/caption of the whole table.	Centered above the table body.	Must be the immediate first child of <table>.
<thead>	Groups header row(s) for the table body.	Block grouping container; no visual change.	Placed before <tbody> and <tfoot>, enclosing <tr>.
<th>	Defines a specific header cell within a row.	Bold text, centered horizontally.	Placed inside <tr> rows (typically inside <thead>).

Question 101: Write HTML code to show each letter of word RAINBOW with new color, giving a rainbow effect.

Key Concept & Explanation: In pure HTML (without CSS), letter colors are achieved using the native HTML tag across the 7 VIBGYOR colors:

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Rainbow Effect - Pure HTML</title>
</head>
<body>
  <h2>Rainbow Text (Pure HTML Using Font Tag):</h2>

  <h1>
    <font color="violet">R</font>
    <font color="indigo">A</font>
    <font color="blue">I</font>
    <font color="green">N</font>
    <font color="yellow">B</font>
    <font color="orange">O</font>
    <font color="red">W</font>
  </h1>
</body>
</html>
```

Question 102: How <meta> tag is helpful for Search Engine Optimization(SEO)?

Key Concept & Explanation: SEO functions provided by HTML <meta> tags placed inside <head>:

- 1. Meta Description (<meta name='description' content='...>):** Supplies search engines with a clear synopsis displayed in the search results snippet. An engaging description boosts organic click-through rates (CTR).
- 2. Responsive Viewport (<meta name='viewport' content='width=device-width, initial-scale=1.0>):** Signals to Google that the webpage is formatted for mobile devices, which is critical for mobile-first indexing rankings.
- 3. Robots Directives (<meta name='robots' content='index, follow>):** Instructs web spiders whether to crawl and index the web page content and links.
- 4. Meta Charset (<meta charset='UTF-8'>):** Prevents character decoding errors across multiple language queries.

Pure HTML Head Section with SEO Meta Tags:

```
<!DOCTYPE html>
<html>
<head>
  <!-- Essential Pure HTML SEO Meta Tags -->
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Full Stack Web Development Notes</title>
  <meta name="description" content="Complete university exam notes and HTML solutions for Semester III Full Stack Development.">
  <meta name="keywords" content="HTML, Web Development, GTU, Semester III">
  <meta name="author" content="Priyen Patel">

</head>
<body>
  <h1>Welcome to Full Stack Web Development</h1>
</body>
</html>
```

Question 103: How to display ordered list and unordered list using HTML? Give Example.

Key Concept & Explanation: Pure HTML uses for numbered/alphabetical sequences and for bulleted items, using the native 'type' attribute without any CSS.

Pure HTML Solution Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Ordered and Unordered Lists</title>
</head>
<body>
  <!-- Ordered List (Numbered) -->
  <h2>Steps in Web Development (Ordered List)</h2>
  <ol type="1">
    <li>Design HTML page structure</li>
    <li>Implement server logic</li>
    <li>Connect relational database</li>
    <li>Deploy to web server</li>
  </ol>
```

```

<!-- Unordered List (Bulleted) -->
<h2>Frontend Languages (Unordered List)</h2>
<ul type="disc">
  <li>HTML (Structure)</li>
  <li>CSS (Styling)</li>
  <li>JavaScript (Behavior)</li>
</ul>
</body>
</html>

```

Question 104: Create HTML Page for following features: 1) Use various bullet styles 2) Created nested lists

Key Concept & Explanation: Pure HTML supports bullet types directly using the 'type' attribute on (disc, circle, square) and on (1, A, a, I, i). Nested lists are built by nesting another list inside an element.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Bullet Styles and Nested Lists</title>
</head>
<body>
  <h2>Feature 1: Various Bullet Styles</h2>

  <!-- Disc (Default) -->
  <ul type="disc">
    <li>Disc Bullet Item 1</li>
    <li>Disc Bullet Item 2</li>
  </ul>

  <!-- Circle -->
  <ul type="circle">
    <li>Circle Bullet Item 1</li>
    <li>Circle Bullet Item 2</li>
  </ul>

  <!-- Square -->
  <ul type="square">
    <li>Square Bullet Item 1</li>
    <li>Square Bullet Item 2</li>
  </ul>

  <hr>

  <h2>Feature 2: Nested Lists</h2>
  <ol type="1">
    <li>Computer Engineering Subjects
      <ul type="circle">
        <li>Database Management Systems
          <ol type="A">

```



```

<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>HTML Layout Template</title>
</head>
<body>
  <header>
    <h1>Department of Computer Engineering</h1>
  </header>

  <nav>
    <a href="#home">Home</a> |
    <a href="#about">About Us</a> |
    <a href="#contact">Contact</a>
  </nav>

  <main>
    <article>
      <h2>Full Stack Development - Semester III</h2>
      <p>Welcome to Unit 2 covering HTML elements, tables, lists, and forms.</p>
    </article>

    <aside>
      <h3>Notice Board</h3>
      <p>Mid-semester examinations commence shortly.</p>
    </aside>
  </main>

  <footer>
    <p align="center">&copy; 2026 Engineering College. All Rights Reserved.</p>
  </footer>
</body>
</html>

```

Question 106: Write an HTML script to code following table:

Key Concept & Explanation: Pure HTML table using native attributes: 'border', 'cellpadding', 'cellspacing', 'align', 'rowspan', and 'colspan' without any CSS.

Pure HTML Solution Code:

```

<!DOCTYPE html>
<html>
<head>
  <title>Time Table Food Menu</title>
</head>
<body>
  <table border="1" cellpadding="10" cellspacing="0" width="500" align="center">
    <caption><b><font size="5">Time Table</font></b></caption>

```

```

<!-- Row 1: Headers -->
<tr align="center">
  <th colspan="3">Header 1</th>
  <th colspan="2">Header 2</th>
</tr>

<!-- Row 2: Food Items -->
<tr align="center">
  <td rowspan="3"><b>Food Items</b></td>
  <td>Pizza</td>
  <td>Burger</td>
  <td colspan="2">Fries</td>
</tr>

<!-- Row 3 -->
<tr align="center">
  <td>Maggie</td>
  <td>Pasta</td>
  <td>Rolls</td>
  <td rowspan="2">Sticks</td>
</tr>

<!-- Row 4 -->
<tr align="center">
  <td colspan="2">Grill</td>
  <td>Non-Grill</td>
</tr>

<!-- Row 5: Cold Drinks -->
<tr align="center">
  <td><b>Cold Drinks</b></td>
  <td colspan="4">Coca Cola</td>
</tr>

<!-- Row 6: Confirm Order -->
<tr align="center">
  <td colspan="5"><b>Confirm Order</b></td>
</tr>
</table>
</body>
</html>

```

Question 107: Write an HTML code to create the table as shown in the figure given alongside.

Key Concept & Explanation: Created strictly using pure HTML attributes: 'border', 'cellpadding', 'cellspacing', 'align', 'rowspan', 'colspan', nested <ol type='A'>, <ol type='I'>, and <ul type='circle'> without any CSS.

Pure HTML Solution Code:

```

<!DOCTYPE html>

<html>

```

```

<head>

    <title>Product Catalog</title>
</head>

<body>

    <table border="1" cellpadding="8" cellspacing="0" width="650" align="center">

        <!-- Header Row -->
        <tr align="center">

            <th>Category</th>

            <th>Product</th>

            <th>Details</th>

            <th>Quantity</th>

            <th>Price</th>

        </tr>


        <!-- Electronics Category -->
        <tr align="center">

            <td rowspan="3"><b>Electronics</b></td>

            <td rowspan="2">Laptop</td>

            <td align="left">

                <!-- Details with nested lists in pure HTML -->
                <ol type="A" start="5">

                    <li>Brand A

                        <ol type="I" start="9" reversed>

                            <li>Intel i5</li>

                            <li>8GB RAM</li>

                        </ol>

                    </li>

                    <li>Brand B

                        <ul type="circle">

                            <li>Intel i7</li>

                            <li>16GB RAM</li>

                        </ul>

                    </li>

                </ol>

            </td>

        </tr>
    </table>

```

```

        </ol>

    </td>

    <td>5</td>

    <td>$500</td>

</tr>

<tr align="center">
    <td>3</td>
    <td>$600</td>
    <td></td>
</tr>

<tr align="center">
    <td>Smartphone</td>
    <td colspan="2">Brand C</td>
    <td>$300</td>
</tr>

<!-- Accessories Category -->
<tr align="center">
    <td rowspan="2"><b>Accessories</b></td>
    <td>Mouse & Keyboard</td>
    <td colspan="2">Wireless</td>
    <td>$50</td>
</tr>

<tr align="center">
    <td>Headphones</td>
    <td colspan="2">Over-ear, Noise-cancelling</td>
    <td>$80</td>
</tr>
</table>
</body>

```

</html>